2. Identify the like terms from the following:

(i) $2x^2y,\ 2xy^2,\ 3xy^2,\ 14x^2y,\ 7yx$

(ii) $3x^3y^2,\ y^3x,\ y^3x^2,\ -y^3x,\ 3y^3x$

(iii) $11pq,\ -pq,\ 11pqr,\ -11pq,\ pq$

Example 3.14 Find the degree of the following expressions.

(i) x^5 (ii) $-3p^3q^2$ (iii) $-4xy^2z^3$

(iv) $12xyz-3x^3y^2z+z^8$ (v) $3a^3b^4-16c^6+9b^2c^5+7$

N1S4Q4

Solution

(i) In x^5, the exponent is 5. Thus, the degree of the expression is 5.

(ii) In $-3p^3q^2$, the sum of powers of p and q is 5 (that is, 3+2). Thus, the degree of the expression is 5.

(iii) In $-4xy^2z^3$, the sum of powers of x, y and z is 6 (that is, 1+2+3). Thus, the degree of the expression is 6.

(iv) The terms of the given expression are $12xyz,\ 3x^3y^2z,\ z^8$

Degree of each of the terms: 3, 6, 8

Terms with highest degree: z^8.

Therefore, degree of the expression is 8.

(v) The terms of the given expression are $3a^3b^4, -16c^6,\ 9b^2c^5,\ 7$

Degree of each of the terms: 7, 6, 7, 0

Terms with highest degree: $3a^3b^4, 9b^2c^5$

Therefore, degree of the expression is 7.

Example 3.15 Add the expressions $4x^2+3xy+9y^2$ and $2x^2-9xy+6y^2$ and find the degree.

Solution

This can be written as $(4x^2+3xy+9y^2)+(2x^2-9xy+6y^2)$

Let us group the like terms, thus we have

$$\left(4x^2+2x^2\right)+\left(3xy-9xy\right)+\left(9y^2+6y^2\right) = x^2\left(4+2\right)+xy\left(3-9\right)+y^2\left(9+6\right)$$
$$= 6x^2-6xy+15y^2$$

Thus, the degree of the expression is 2.

Example 3.16 Subtract x^3-x^2+x+3 from $3x^3-2x^2-7x+6$ and find the degree.

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